Luis Henrique Lima Santos

João Pessoa, PB, Brazil

henrique.santos.lhls@gmail.com | +55 83 986630526 | github.com/henrique-lh | linkedin.com/in/henrique-lh

SUMMARY

Software Engineer specializing in MLOps, backend systems, and production ML infrastructure. Experienced in Kubernetes-based model deployment (KServe), CI/CD automation with GitHub Actions, and scalable backend services using FastAPI and ASP.NET (F#). Strong background in distributed systems, model monitoring, and infrastructure automation.

EXPERIENCE

Datarisk Aug. 2025 – Present

Software Engineer Brazil (Remote)

- Contributed to a distributed architecture integrating React (TypeScript), F# (Giraffe/ASP.NET), RabbitMQ, and Apache Airflow to support asynchronous ML workloads and real-time communication via WebSockets.
- Architected an automated alert system (Slack/Email) to monitor model performance and token validity, reducing manual oversight hours by 99% weekly and preventing critical downtime.
- Migrated 15+ containerized ML services to KServe on Kubernetes, enabling autoscaling and scale-to-zero model deployment, reducing operational overhead by 40%.
- Developed CI/CD pipelines with GitHub Actions to build Docker images, publish versioned Python packages, and automate testing and deployment workflows.
- Implemented model drift detection dashboards including SHAP analysis, statistical distribution monitoring, and central tendency metrics persisted in PostgreSQL.
- Contributed to the open-source `datarisk-mlops-codex` library, improving reliability, troubleshooting production issues, and supporting client adoption.

Datarisk Mar. 2024 – Aug. 2025

Trainee/MLOps Intern Florianópolis, Brasil

- Designed automated ML monitoring pipelines using Airflow, DuckDB, and NannyML to evaluate regression (DLE, MAPE) and classification (CBPE, ROC, AUC) models in production, enabling proactive retraining decisions across 15+ models production models.
- Developed Airflow DAGs to compute statistical monitoring metrics (mean, median, mode) from training and production datasets, persisting results in PostgreSQL for traceability.
- Built a secure preprocessing pipeline for controlled execution of usersubmitted scripts via REST APIs, ensuring validation and isolation in a multitenant environment.

Enacom Group Aug. 2023 – Jan. 2024

Junior Software Developer Belo Horizonte, Brasil

- Developed a constraint-based simulation system for rail and port logistics operations at VLI Logística, modeling transportation capacity and product flow (sugar and fuel) across railways, highways, and ports. Implemented mathematical models to optimize tonnage allocation under operational constraints, supporting large-scale transportation planning decisions.

PROJECTS

Genetic Syndrome Classification | Python, TensorFlow, FastAPI

- Developed a machine learning pipeline to classify genetic syndromes from facial image embeddings using deep learning models.
- Built a FastAPI inference API for real-time predictions and modular preprocessing pipeline for feature extraction.

Investment Selection Optimization Tool | Python, Pandas, NumPy

- Engineered a portfolio optimization system using Knapsack dynamic programming to maximize returns under budget constraints.
- Reduced computational complexity from $O(2^n)$ to O(n x W), decreasing processing time by 95% on large datasets.

Agricultural Product Evaluation System | YOLO, Raspberry Pi, Python

- Built a real-time agricultural monitoring system integrating IoT hardware and YOLO-based object detection for automated product sorting.
- Implemented computer vision inference pipeline deployed on edge hardware (Raspberry Pi).

EDUCATION

Instituto Federal da Paraíba (IFPB) Jan. 2020 – Apr. 2025
Bachelor of Engineering, Computer Engineering – GPA: 9.0/10 Campina Grande, PB, Brazil

- **Coursework**: Data Structures, Algorithms, Databases, Computer Systems, Machine Learning
- **Research Fellow (May 2023 – Dec 2023)**: Developed a machine learning-based risk impact prediction system for software projects, implementing data analysis pipelines in Python and WEKA to estimate project success probability.
- **Publication**: Co-authored peer-reviewed paper in SBC proceedings on software engineering risk analysis.
- **Teaching & Mentorship (2021)**: Instructor Scholar for Programming Olympics and Teaching Assistant for Data Structures & Algorithms, covering graphs, trees, dynamic programming, and divide-and-conquer techniques.

SKILLS

- **Programming Languages**: Python, TypeScript, JavaScript, F#, SQL
- **Backend & APIs**: FastAPI, ASP.NET Core, Giraffe, REST APIs, WebSockets, Microservices Architecture
- **Machine Learning & Data**: Scikit-learn, TensorFlow, Keras, NannyML, Airflow, DuckDB, Pandas, NumPy, SHAP, Model Monitoring, Drift Detection
- **DevOps & MLOps**: Docker, Kubernetes, KServe, GitHub Actions, CI/CD, Terraform, AWS, Containerization, Infrastructure as Code
- **Databases & Messaging**: PostgreSQL, MongoDB, Redis, RabbitMQ
- **Tools**: Git, Linux
- **Soft skills**: Teamwork (Waterfall, Agile, Scrum), Interpersonal Skills, Proactivity, Attention to Detail, Problem Solving, Critical & Creative Thinking, Resilience, Work Ethic, Time Management.